

Rate of turn, turn radius and the steering linkage — Northstar 76, 2026 season

26 September 2026. Every number is measured from the boat's own logs. Where the logs cannot support a measurement it is left out rather than estimated, and the coverage is stated beside each figure.

1. The three numbers, and why they differ

Rate of turn — heading change over the **6 seconds centred on the moment the apparent wind crosses the bow**, divided by the span actually measured. The average through the fastest part of the turn.

Turn radius — **speed ÷ yaw rate** over that same window. The radius the boat actually turned at. This is the operating point any steering geometry has to be correct for, and unlike the geometry it can be measured without knowing anything about the boat.

Max rotation — the single fastest heading change between two consecutive samples. A *peak*, which rises with the log's own sampling noise.

Compared **on the same manoeuvres**, max rotation reads higher in **94 of 98 tacks** and **44 of 44 gybes**, by a median of **1.00 °/s** and **1.19 °/s**. That gap is the sampling, not the sailing. A target set from max rotation would be a target nobody is actually sailing.

Timing the window on the apparent wind rather than on the manoeuvre detector matters: on 12 September the crossing lands a consistent **+2.7 s after** the detected moment, so timing from the detector would have measured a different part of every turn.

What the season can carry

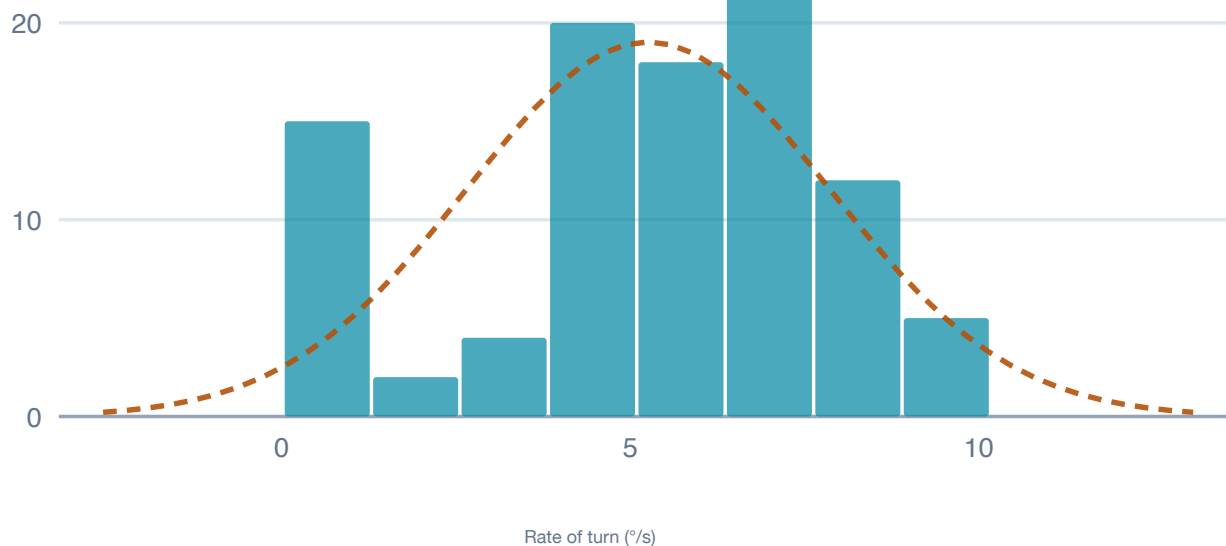
	tacks	gybes
Rate of turn (±3 s window)	98 of 587, on 8 of 23 days	44 of 230, on 6 of 22 days
Turn radius	81	42
Max rotation	546	220

Rate of turn needs a log sampled at **3 s or finer**; half the season is logged at 6.09 s, and a six-second window sampled every six seconds is not a measurement. Turn radius needs the rate *and* a real turn under it — **$R = v/\omega$ runs away as ω goes to zero**, so it is only computed above 2 °/s. Max rotation survives a coarse log, which is why it covers five times as many manoeuvres, and why it should be read as a ceiling rather than a performance.

2. Tacks

Rate of turn across 98 tacks

■ Rate of turn · 98



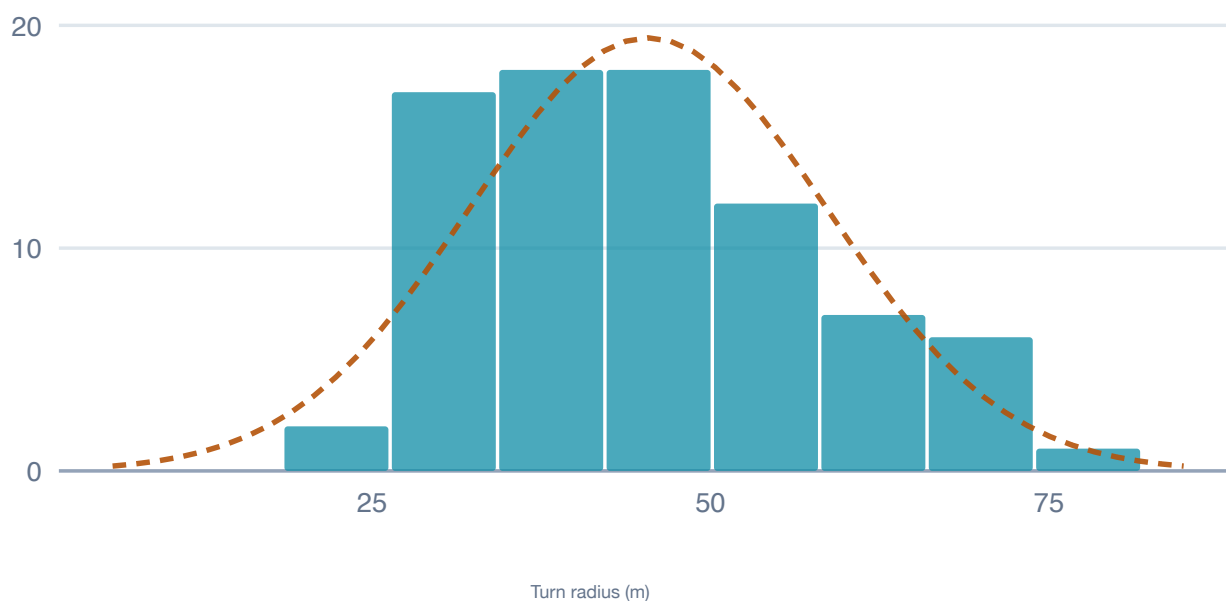
Rate of turn through 98 tacks. Bars are counts; the dashed line is the normal fitted to the same mean and spread, drawn so the difference from it can be seen.

It is not a bell. A hump at 5–7 °/s and a tail of slow ones running to nearly zero. The mean of **5.25 °/s** sits below the median of **5.73** because that tail drags it there; the typical tack is the median.

The bottom bar is not slow tacking. **Fifteen of the ninety-eight** sit near zero, and a tack that turns at 0.3 °/s is something recorded as a tack that was not one, or a stretch where the heading data is bad. They are excluded from the radius for that reason.

Turn radius across 81 tacks

■ Turn radius · 81

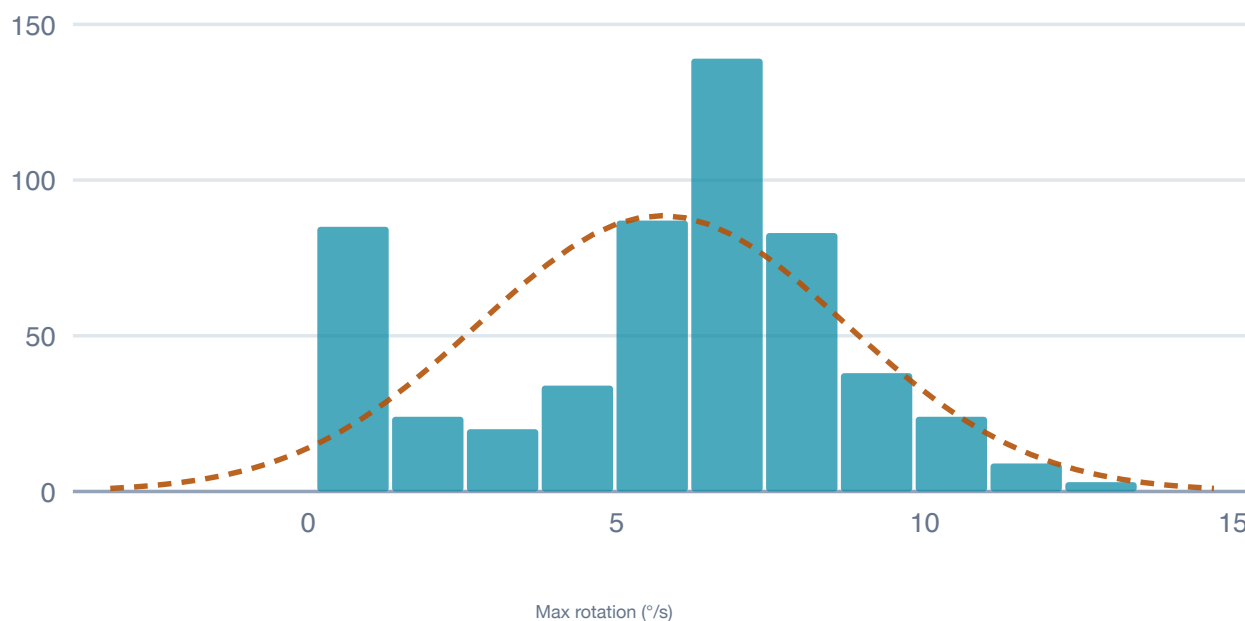


Turn radius through 81 tacks, in metres.

The boat tacks in about 43 m — median, against a hull length of 23.16 m, so **1.86 boat lengths**, and tightly grouped: 18 m to 82 m with a spread of 13 m.

Max rotation across 546 tacks

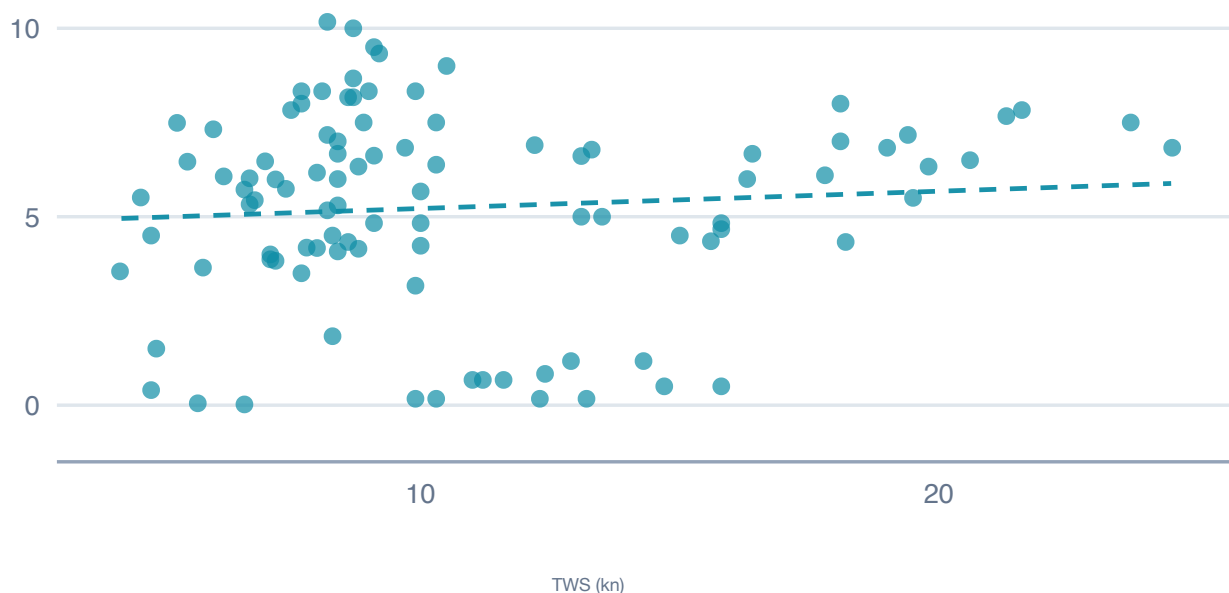
■ Max rotation · 546



Max rotation through 546 tacks — five times the sample, because a coarse log still supports it.

Rate of turn vs TWS (°/s)

● Tacks · 98 · R^2 0.01

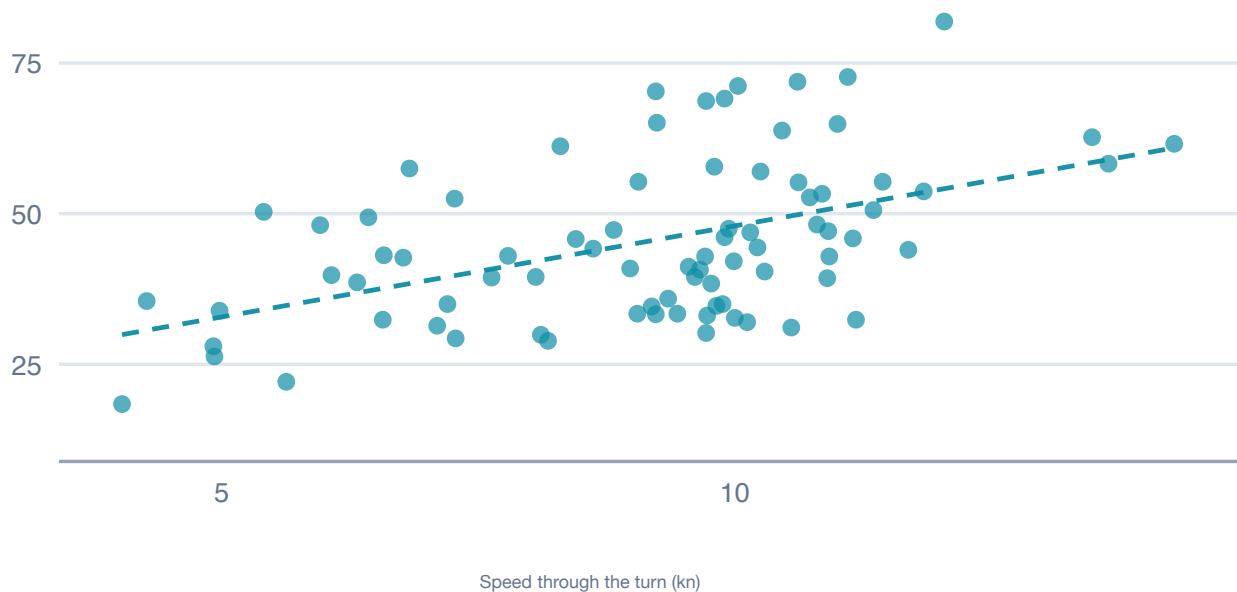


Rate of turn against true wind speed, 98 tacks, with a least-squares line.

Rate of turn does not depend on the breeze. R^2 is **0.01** across 7–20 knots. Whatever governs how fast this boat comes through the wind, it is not wind strength — so it is worth looking at what the crew does rather than at conditions.

Turn radius vs Speed through the turn (m)

● Tacks · 81 · R^2 0.24

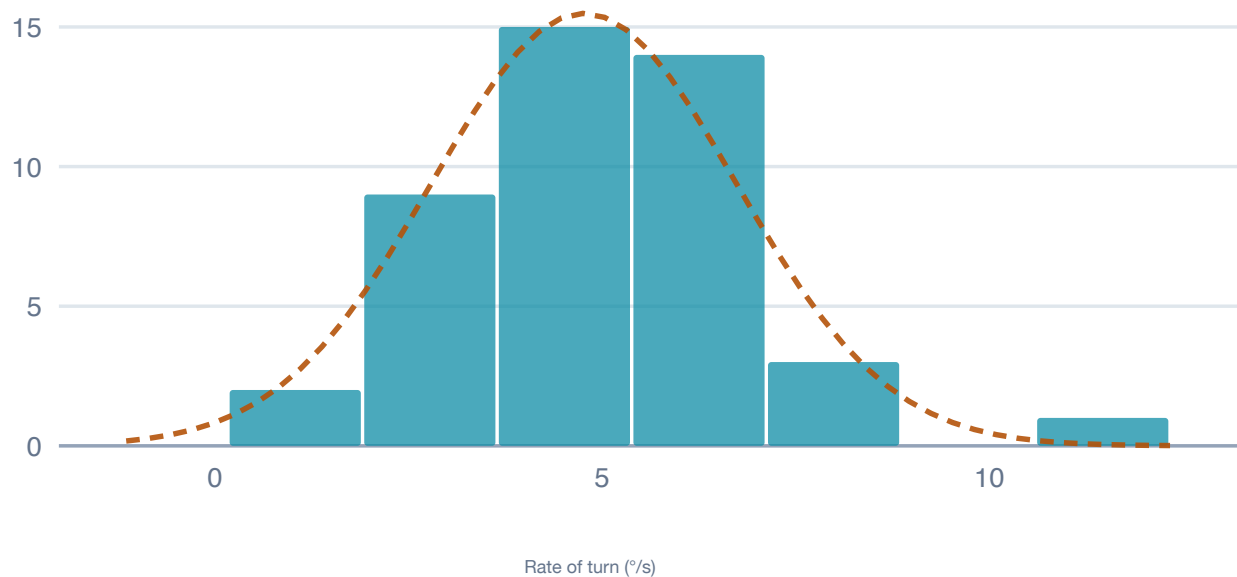


Turn radius against the speed carried through the turn.

3. Gybes

Rate of turn across 44 gybes

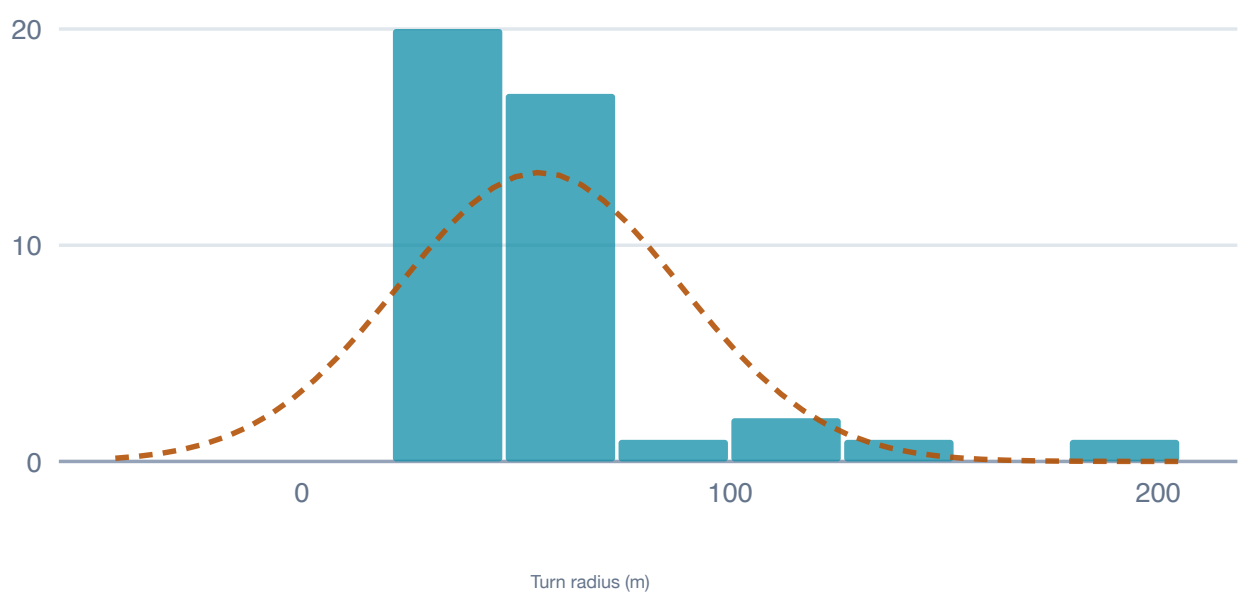
■ Rate of turn · 44



Rate of turn through 44 gybes.

Turn radius across 42 gybes

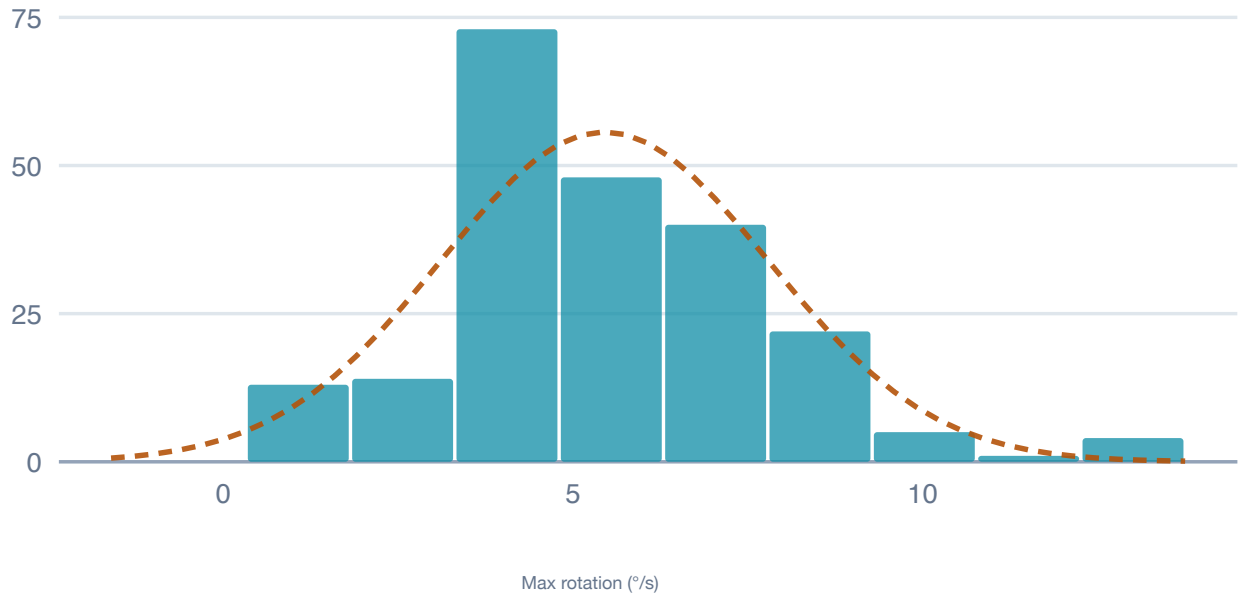
■ Turn radius · 42



Turn radius through 42 gybes.

Max rotation across 220 gybes

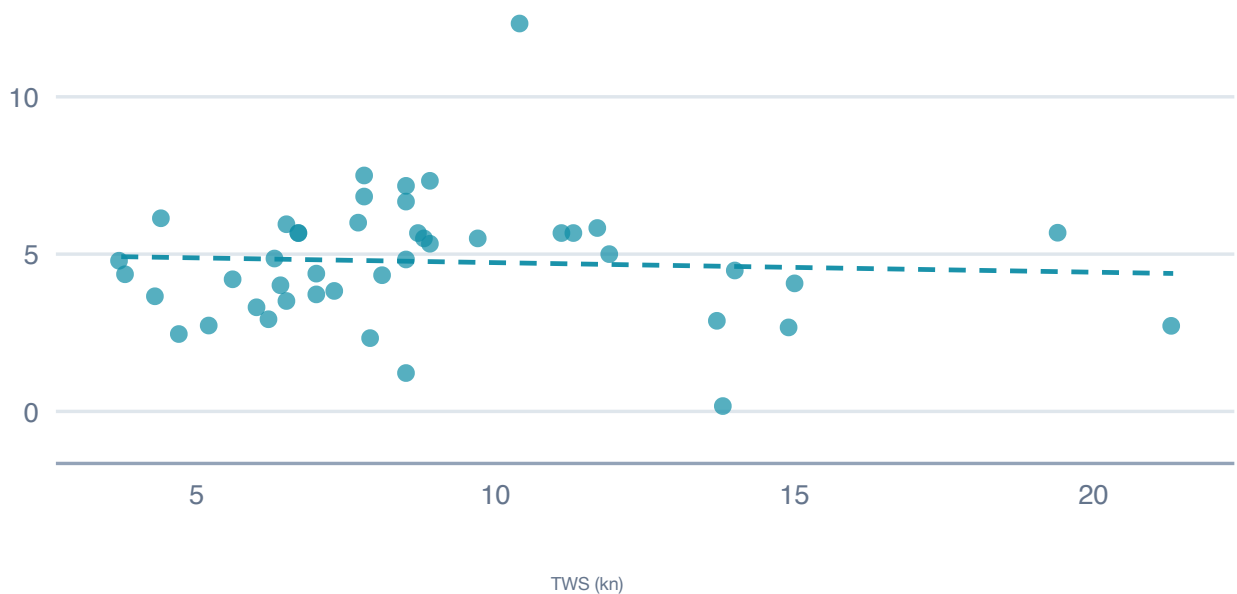
■ Max rotation · 220



Max rotation through 220 gybes.

Rate of turn vs TWS (°/s)

● Gybes · 44 · R^2 0.00



Rate of turn against true wind speed, 44 gybes.

Gybes turn **slower and wider** than tacks — 4.81 °/s against 5.73, and a radius of **49.6 m (2.15 boat lengths)** against 43.0 — but the rate is more tightly grouped (spread 1.97 against 2.61). A gybe is a smaller heading change made under more control, and it is the more repeatable manoeuvre on this boat. The radius spread is the wider of the two (33 m against 13 m), driven by a handful of very wide ones out to 205 m.

4. Summary

- The typical tack turns at **5.7 °/s** in **43 m**; the typical gybe at **4.8 °/s** in **50 m**. Medians, from the ± 3 s window.
- The boat turns in **roughly two boat lengths**, and tacks are the tighter of the two manoeuvres.
- The tack rate distribution is **not normal** — a hump with a tail of slow ones. The tail is where the time goes, not the average.
- **Rate of turn is independent of wind speed** (R^2 0.01). If it is to be improved, the variable is not the breeze.
- **Fifteen of the ninety-eight rated tacks look like bad records rather than bad tacks** and should be checked before any mean is quoted onward.
- Max rotation reads higher on the same manoeuvre almost every time. Use it for coverage, not as a target.

5. Steering geometry

Everything in this section is built on **judged manoeuvres carrying a measured turn direction** — 33 tacks and 13 gybes. That is a smaller population than the distributions in sections 1–3, which is deliberate: [isJudged](#) drops mark roundings and pre-start work, and a manoeuvre with no direction cannot say which rudder was on the inside of the turn, which is the whole question here.

The sign convention, established rather than assumed

An earlier version of this section had the inner rudder **under**-turned by 4.6° and recommended moving the slider 1770 mm. That was wrong, and the error was one assumption: that a positive logged rudder angle meant turning to starboard.

It means the opposite. Checked against heading change across every turning row in the season, **a positive logged rudder angle goes with a decreasing heading in 7,053 of 7,347 rows — 96%. **Positive helm is a turn to port****. Inner and outer were therefore swapped, and the sign of the conclusion with them.

The raw data shows it without any modelling, on 12 September:

heading	turn	ruddP	ruddS	inner
311 → 345	starboard	−12.68°	−16.62°	starboard, by 3.94° more
191 → 135	port	+16.94°	+14.16°	port, by 2.78° more

The inner rudder carries more angle in both directions. The boat is running Ackermann in the correct sense. The only question is how much.

The same trap is worth naming, because it is easy to walk into again: the mean of `ruddS − ruddP` pooled over both directions is **not** a measure of Ackermann. Which rudder is inner flips with the turn, so pooling the two directions measures the *offset* and hides the differential. It has to be split by direction, and the angles compared as magnitudes.

Do the two rudders match what the turns required?

Each manoeuvre compared against the ideal for **its own** radius, then the median of those errors — not a median against a median, which hides the spread.

Tacks — 33 manoeuvres (16 to starboard, 17 to port)

	inner	outer	difference
Turn radius	44.0 m from BSP · 45.3 m from SOG (103%)		
Effective pivot distance L	16.1 m		
Required (ideal Ackermann)	19.40°	17.65°	+1.76°
Measured	20.07°	16.03°	+4.49°
Correction , raw			−2.62°
Correction , sensor offset removed			−1.50°

Gybes — 13 manoeuvres (4 to starboard, 9 to port)

	inner	outer	difference
Turn radius	49.8 m from BSP · 53.0 m from SOG (106%)		
Effective pivot distance L	11.6 m		
Required (ideal Ackermann)	13.70°	12.56°	+1.14°
Measured	14.98°	11.69°	+3.11°
Correction , raw			−1.26°
Correction , sensor offset removed			−0.14°

The boat runs slightly too much Ackermann, not too little — by a tenth of a degree to two and a half, depending on the manoeuvre and on how the constant offset is read. Not the 4.6° reversal the earlier draft reported.

Two things make the radius trustworthy. Speed through the water and speed over the ground give radii within **3% on tacks and 6% on gybes**, so the number is not an artefact of which speed was used. And the error is **robust to L**: across a pivot distance of 10 m to 15 m the tack error only moves from +1.89° to +1.47°.

Thirteen gybes, only four of them to starboard, is thin. Treat the gybe table as indicative.

The asymmetry is larger than the error being corrected

tacks	radius	inner	outer	difference
turning to starboard (n=16)	41.7 m	22.55°	16.61°	+6.16°
turning to port (n=17)	46.1 m	19.40°	15.87°	+3.70°

A sensor zero cannot produce this. A constant offset adds the same **+1.10°** to inner-minus-outer in *both* directions — the algebra works out the same way whether the helm is positive or negative, because the inner rudder swaps with it. So a 2.5° gap between the two directions is something else: an asymmetric linkage as built, different gains on the two sensors, or the boat genuinely being tacked differently each way. It is worth settling before chasing a 1.5° error, because it is bigger.

The linkage, rebuilt

The linkage figures in the earlier draft do not close geometrically — a 349 mm tiller on a stock 2238 mm off the centreline cannot produce a 1761 mm rod to a slider 679 mm aft. Those numbers came from the superseded 4.0 m-separation model and are withdrawn.

Modelled from the dimensions that are known: **stocks 4476 mm apart, 349 mm tiller arms, two rods to one slider on the centreline** which translates athwartships as the steering input. Two unknowns remain — the slider's fore-and-aft offset from the line of the stocks, and the neutral angle of the tiller arms — so rather than guess them, the offset is **fitted to the measured differential** for each plausible arm angle:

neutral arm angle	slider offset	rod length	differential delivered
−10°	565 mm forward of the stocks	2472 mm	4.49°
−5°	775 mm forward	2531 mm	4.49°
0° (arms straight aft)	1005 mm forward	2616 mm	4.49°
+5°	1270 mm forward	2737 mm	4.50°
+10°	1570 mm forward	2899 mm	4.48°

Any of these reproduces what the boat does. **The useful finding is that the answers below are the same across all of them**, so they do not depend on which is the real build.

Shortening the tillers by 20 mm

349 mm → 329 mm, both sides, nothing else changed:

	349 mm	329 mm	change
Slider travel for 20.07° of helm	116.9 mm	110.3 mm	−5.7%
Helm at the same slider stroke	20.07°	21.41°	+1.34°, +6.7%
Ackermann differential at 20.07° helm	4.49°	4.42°	−0.07°

(Figures for arms straight aft; across the whole family above the gain change stays −5.5% to −5.8% and the differential change −0.06° to −0.07°.)

Shortening the tillers is a gearing change, not an Ackermann change. It buys about **6.5% more rudder angle for the same wheel movement** — the same helm for 5.7% less slider travel — and the steering will load up correspondingly, because the same rudder torque now acts on a 5.7% shorter lever.

Its effect on the differential is **0.07°, against an excess of 1.50°**. Twenty millimetres of tiller buys under 5% of the correction. Extending the reasoning: the differential moves about **0.065° per 20 mm** of tiller, so removing 1.50° by tiller length alone would need roughly **460 mm** off a 349 mm arm. It is not available.

Tiller length and Ackermann are very nearly orthogonal controls. That is a useful thing to know rather than a disappointment: the tillers can be shortened for the gearing the helm wants, without disturbing the steering geometry.

What the Ackermann control actually is

The slider's fore-and-aft offset, at about **0.35° of differential per 100 mm**, and it moves the differential *down* as the slider goes aft:

if the 1.10° offset is...	the linkage should deliver	slider move needed
a sensor calibration error	2.98° at 20.07° helm	405–485 mm aft
real built-in toe	1.88° at 20.07° helm	715–835 mm aft

The range across the family of arm angles is the width of those brackets. Which row applies is decided by one dockside measurement, and the two differ by 300 mm of slider position — which is why that measurement comes first.

In order

1. **Zero both rudder sensors** dockside, against physically aligned rudders. This decides which row of the table above applies, and may account for the whole remaining error.
2. **Establish where the port/starboard asymmetry comes from** — compare the two rod lengths and the two tiller arms as built, and check the sensors against a protractor at matched angles each side. At 2.5° it is the largest single term.
3. **Read off the slider ball joint's x, y, z and its travel axis** from the CAD, and **measure L** by tape. The model fits the slider offset because it is unknown; one measurement replaces the whole family above with a single geometry.
4. **Shorten the tillers if the gearing is wanted** — that decision is independent of everything above, and the 20 mm is worth about 6.5% more helm per turn of the wheel.
5. **Only then move the slider**, and re-measure on a 1–2 s log.

Generated from `session_phase_stats` and the session logs, 26 September 2026. Rate of turn: heading change over the 6 s centred on apparent wind = 0, on logs of 3 s or finer, gated on the crossing being interpolated across ≤ 3 s and the window holding ≥ 3 headings. It is a magnitude; the turn DIRECTION is recorded separately, from a signed heading wrap, because which rudder was on the inside of the turn cannot be recovered from the rate alone. Turn radius: mean speed over the same window divided by the yaw rate, above 2 °/s only, cross-checked against the same calculation from SOG. Rudder sign convention established empirically, not assumed: a positive logged angle accompanies a decreasing heading in 7,053 of 7,347 turning rows across the season (96%), so positive helm is a turn to port. Inner and outer are assigned per manoeuvre from its own heading change and compared as magnitudes, and each manoeuvre is measured against the ideal for its own radius before the median is taken. Section 5 uses judged manoeuvres carrying a direction: 33 tacks, 13 gybes. Effective pivot distance: $R \cdot \tan \delta$ with δ the mean helm over the turn. Linkage: two rods from 349 mm tiller arms to one centreline slider translating athwartships, stocks 4476 mm apart, solved numerically. The slider's fore-and-aft offset and the arms' neutral angle are not known, so the offset is fitted to the measured differential across a family of arm angles; every conclusion drawn from it is reported across that whole family rather than from one member of it.